EduManage

Complete Student Management & Educational ERP System

Executive Project Summary:
Yeh document **EduManage (Student Management System)** ka complete technical, architectural aur operational report hai. Isme explain kiya gaya hai ki is project me **kya-kya banaya gaya**, **kaun-kaun si technologies use hui**, **system kaise build kiya gaya**, database schema aur data models kaise connect hain, aur internal data flow browser se lekar database tak kaise execute hota hai.

Project Category	Full-Stack Web ERP / Academic Institutional System
Backend Core	Python 3.x, Flask Web Framework (WSGI, Jinja2, Werkzeug)
Database	SQLite3 (Embedded Relational DB, 7 Interlinked Tables)
Frontend UI/UX	HTML5, Vanilla CSS3 (Custom Glassmorphism), JavaScript (DOM Live Search)
Security & Auth	PBKDF2 Password Hashing, Session Security, @login_required Route Decorator
Modules Built	Auth & Passwords, KPI Dashboard, Student CRUD, Courses, Subjects, Attendance, Exams/Marks, Fees, 360° Profile
Project Status	**100% Operational & Production Ready (Phase 4 Completed)**

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1. Executive Project Overview (■■■■■■■■■ ■■ ■■■■■ ■■ ■■■■■■■■)

EduManage ek full-fledged, modular aur secure **Student Management Web Application** hai. Educational institutions (jaise colleges, institutes, aur schools) me student records ko maintain karne ke liye pehle manual files, physical registers ya disconnected Excel spreadsheets use hoti thi. Isse human errors, duplicate roll numbers, attendance register gum hone, aur fee dues miss hone ki badi samasya aati thi. Is project ko ek comprehensive, single-source-of-truth ke roop me build kiya gaya hai jo student life-cycle ke sabhi stages (Admission -> Course/Subject Mapping -> Daily Attendance -> Examination Marks -> Fee Collection -> Academic Transcript) ko automate karta hai.

Mukhya Lakshya (Key Objectives):

- **Centralized Student Repository:** Sabhi students ka personal, contact aur academic record ek structured database me secure rakhna.
- **Integrity & Zero Duplication:** Unique roll numbers enforce karna taaki duplicate admission bilkul na ho sake.
- **Daily Attendance Automation:** Single-click interface se daily attendance record karna aur percentage automatically calculate karna.
- **Examination Automation:** Subject-wise marks save hote hi overall aggregate percentage automatic recalculate hona.
- **Financial Accountability:** Transparent fee collection ledger maintain karna jisme total fee, paid amount aur pending balance real-time track ho.
- **Access Security:** Hashed passwords aur session protection ke zariye administrative data ko unauthorized access se bachana.

2. Technology Stack & Tools Used (■■■-■■■ ■■ ■■■■■■■■■■■ ■■■■■■■■ ■■■■)

Project ko lightweight, fast, modern aur production-ready banane ke liye standard open-source technologies ka use kiya gaya hai:

Layer	Technology / Tool	Role & Implementation Details
Backend Core	**Python 3.x**	Core programming language; application logic, business rules, mathematical formulas, aur SQLite connectivity handle karta hai.
Web Framework	**Flask 3.x**	Micro web framework jo HTTP routes, request processing, sessions, URL dispatching, aur Jinja2 template rendering handle karta hai.
Database	**SQLite 3**	Serverless, zero-configuration relational database (database.db). Relational foreign keys, ACID compliance aur instant queries provide karta hai.
Template Engine	**Jinja 2**	Dynamic server-side templating; master template inheritance (base.html), conditional statements, loops aur context passing provide karta hai.
Frontend Styling	**Vanilla CSS3**	Custom Glassmorphism UI tokens, frosted glass cards (backdrop-filter), modern color variables, CSS Grid aur Flexbox layouts.
Client Scripting	**Vanilla JavaScript**	DOM manipulation aur event listeners for real-time table filtering without page reloads.
Security & Auth	**Werkzeug Security**	Cryptographic PBKDF2/scrypt password hashing (generate_password_hash, check_password_hash) aur signed cookie sessions.
Icons UI	**Boxicons CDN**	Vector icons for sidebar navigation, KPI cards, student actions, and status badges.
Report Engine	**ReportLab 5.x**	Programmatic PDF generation engine used to compile this executive-grade documentation.

3. System Architecture & Design Pattern (■■■■■■ ■■■■ ■■■■■ ■■■)

Is system ko **MVC (Model-View-Controller) / MTV (Model-Template-View)** architectural pattern par develop kiya gaya hai:

- **Model (Data Layer):** SQLite3 tables (students, courses, subjects, attendance, marks, fees, users) jisme data constraints, primary keys aur foreign key relationships enforce hain.
- **Controller / View (Business Logic Layer):** app.py me Flask route handlers aur Python logic jo request validation, authorization checks, calculations (attendance %, pending balance, student GPA average) aur database queries execute karte hain.
- **Template (Presentation Layer):** templates/ directory me Jinja2 templates (jaise base.html) jo dynamic HTML render karte hain, aur static/ directory me CSS styling aur JS behaviors hain.

Project File Structure & Roles (■■■■ ■■■■■■):

File / Folder	Role & Description
`app.py`	Main application backend: Flask app setup, DB connection, @login_required decorator, and 15+ HTTP routes.
`database.db`	SQLite database storing persistent records (students, courses, subjects, attendance, marks, fees, users).
`templates/base.html`	Master layout with glassmorphic sidebar, top notification banners, session user badge, and block contents.
`templates/index.html`	Institute dashboard with KPI stat cards, student table directory, and real-time live search.
`templates/profile.html`	360-degree comprehensive student profile aggregating attendance stats, marksheet, and fee ledger.
`templates/login.html`	Secure administrative login interface with credentials verification and error feedback.
`templates/add_student.html` `templates/edit_student.html`	Student admission and update forms with strict server-side validation (phone, unique roll no, semester, marks).
`templates/attendance.html`	Daily attendance marking portal with date/course/sem filters and daily attendance percentage metrics.
`templates/marks.html`	Examination marks entry sheet with automatic live recalculation of student's overall aggregate score.
`templates/fees.html`	Fee collection accounting dashboard with payment installments, pending dues, and financial summary cards.
`templates/courses.html` `templates/subjects.html`	Academic catalog managers for establishing degree programs and semester-mapped course syllabi.
`static/css/style.css`	Modern Glassmorphism stylesheet with CSS variables, responsive grid/flexbox layouts, cards, and badges.
`static/css/Js/script.js`	Client-side search script for instant row-level DOM filtering without network requests.

4. Relational Database Design & Schema (■■■■■■■ ■■■■■■■ ■■ ■■■■ ■■■■)

System ka database SQLite3 me structured relational schema par organized hai jisme 7 primary tables banaye gaye hain:

Table Name	Primary Columns & Types	Constraints & Relationships
users	id (PK), username (TEXT), password_hash (TEXT), email (TEXT), full_name (TEXT), role (TEXT), created_at (TIMESTAMP)	username UNIQUE, email UNIQUE. Password hashed with PBKDF2 for admin security.

students	id (PK), roll_no (TEXT), name (TEXT), email (TEXT), phone (TEXT), dob (TEXT), gender (TEXT), course (TEXT), semester (INT), marks (REAL), address (TEXT), status (TEXT)	roll_no UNIQUE NOT NULL. Status defaults to 'Active'. Central entity referenced across all modules.
courses	id (PK), name (TEXT)	name UNIQUE NOT NULL. Represents degree programs (B.Tech, BCA, MCA, MBA).
subjects	id (PK), course_id (INT), semester (INT), name (TEXT)	FOREIGN KEY (course_id) REFERENCES courses(id). Maps subjects to course & semester.
attendance	id (PK), student_id (INT), date (TEXT), status (TEXT)	FOREIGN KEY (student_id) REFERENCES students(id). Status: 'Present', 'Absent', 'Late'.
marks	id (PK), student_id (INT), subject_id (INT), marks_obtained (REAL)	FOREIGN KEY (student_id) REFERENCES students(id), FOREIGN KEY (subject_id) REFERENCES subjects(id).
fees	id (PK), student_id (INT), total_fees (REAL), paid_fees (REAL), last_payment_date (TEXT), last_payment_amount (REAL)	student_id UNIQUE NOT NULL. FOREIGN KEY (student_id) REFERENCES students(id). Tracks dues & receipts.

5. Detailed Modules & What Was Built (■■■■■■■■■ ■■■ ■■■■-■■■■ ■■■■■ ■■■)

Is project me complete educational administration ke liye 8 specialized modules build kiye gaye hain:

Module 1: Authentication & Security (Phase 4)

- **Secure Login Portal:** /login route par username aur password authentication. Werkzeug check_password_hash algorithm se verify hota hai.
- **Route Guard Protection:** Custom @login_required decorator jo unauthorized users ko login page par redirect karta hai.
- **Dynamic User Badge:** @app.context_processor decorator template me globally current_user pass karta hai taaki sidebar me logged-in admin ka avatar aur role display ho.
- **Change Password Facility:** /profile/change-password route par admin apna password badal sakta hai jisme current password verification aur minimum 6-character validation enforced hai.
- **Default Admin Seeding:** Database initialization par default admin credentials (admin / admin123) automatically initialize ho jaate hain.

Module 2: Institute KPI Dashboard (Overview Center)

- **Real-time Metrics:** Total Students, Active Students vs Inactive, Total Programs, Total Subjects, aur Institute Aggregate Marks.
- **Student Directory:** Tabular listing jisme student avatar, roll no, program, contact, score, status badge display hota hai.
- **Instant Search Filter:** JavaScript powered live search bar jo bina reload ke name, roll no aur course ko filter karta hai.

Module 3: Student Admission & Full CRUD Lifecycle

- **Student Admission:** Roll no, Name, Email, Phone, DOB, Gender, Course, Semester, Initial Marks, Address input.
- **Strict Validations:** 10-digit numeric phone check, Unique Roll No check, Semester range (1-12), Marks range (0-100).
- **Student Editing & Status Toggle:** Details update karne aur student status ko Active/Inactive mark karne ki suvidha.
- **Secure Record Deletion:** POST request validation ke sath student record ko permanently delete karne ka option.

Module 4: Academic Curriculum (Courses & Subjects)

- **Course Catalog:** Naye courses/degrees add ya remove karna (/courses).

- **Subject Mapping:** Subjects ko particular course aur semester ke sath link karna (/subjects) taaki syllabus structured rahe.

Module 5: Daily Attendance Tracking System

- **Flexible Filtering:** Date selector, Course filter aur Semester filter ke sath selective student attendance list.
- **Radio Status Marking:** Har student ke liye Present, Absent ya Late mark karne ka simple UI.
- **Real-time Daily Stats:** Total marked count, Present percentage (Present + 0.5 * Late formula), Absent count calculate hota hai.
- **Smart Upsert:** Agar us date ka attendance pehle se hai to update hota hai, naya hone par insert hota hai.

Module 6: Marks & Examination Grading System

- **Course & Subject Wise Scoring:** Particular course, semester aur subject select karke students ke marks enter kiye jaate hain.
- **Automated Average Recalculation:** Jaise hi kisi subject ke marks update hote hain, system student ke all-subject marks ka average calculate karke students.marks column me update kar deta hai.

Module 7: Fees Management & Financial Ledger

- **Automatic Student Fee Sync:** Har naye student ke liye default fee (■50,000) account automatic open hota hai.
- **Fee Customization:** Course fee ko student-wise customize karne ka provision.
- **Payment Collection:** Partial installments ya full payment record karna with payment date aur last paid receipt details.
- **Financial Overview Cards:** Total Expected Revenue, Total Collected, Total Outstanding Dues, Fully Paid Students Count.

Module 8: 360-Degree Comprehensive Student Profile

- **Unified Single-Page Dossier (/student/<id>):** Student ki identity, personal info, roll number, course, batch.
- **Attendance Breakdown:** Total classes, Present count, Absent count, Late count, aur visual attendance percentage bar.
- **Academic Transcript:** Semester-wise subjects aur marks obtained ka complete table.
- **Financial Summary:** Total course fee, amount paid, balance remaining, last payment date aur receipt amount.

6. How the System Works & Data Flow (■■■■■ ■■■ ■■■ ■■■■ ■■■ ■■■■ ■■■)

Application ka flow user ke browser request se shuru hokar database persistence aur response presentation tak is tarah chalta hai:

Step	Component	Action & Internal Processing
1	Client Request	User browser me URL enter karta hai ya HTML form submit karta hai (e.g. Admit Student form).
2	Security Guard	Flask routing system aur @login_required decorator check karte hain ki user session valid hai ya nahi.
3	Controller Validation	app.py request payload ko sanitize karta hai: blank fields, roll number duplicates, phone length, semester boundaries check karta hai.
4	Database Transaction	Helper function get_db_connection() SQLite database se connect karta hai aur parameterized queries (?) execute karta hai taaki SQL Injection se protection rahe.
5	Jinja2 Dynamic Render	Controller database rows ko fetch karke Jinja2 templates ko pass karta hai jisme base.html layout extend hokar complete webpage banata hai.

6	Client Enhancement	Browser me CSS styles load hoti hain aur script.js real-time table filtering provide karta hai.

7. Security & Validation Safeguards (■■■■■■■ ■■ ■■■■ ■■■■■■■■■)

System me standard security best practices incorporate kiye gaye hain taaki institutional data safe aur consistent rahe:

- **Cryptographic Hashing:** Passwords kabhi bhi plain text me store nahi hote; Werkzeug ka PBKDF2 algorithm irreversible salt-based hash produce karta hai.
- **SQL Injection Prevention:** Saare database queries parameterized statements (? placeholders) use karte hain, jisse dynamic malicious SQL injection blocks ho jate hain.
- **Encrypted Signed Sessions:** User login state Flask ke cryptographically signed session cookie me secure rehti hai jo app.secret_key se sign hoti hai.
- **Server-Side Strict Validation:** Client-side HTML form ke alawa Python backend me strictly check hota hai ki Roll Number duplicate na ho, phone number 10 digits ka ho, semester 1-12 ho aur marks 0-100 ke bich hon.
- **Flash Messaging System:** User ko har action par feedback milta hai (e.g. green success message ya red error alert).

8. How to Setup & Run the Application (■■■■■■■■■ ■■■■■ ■■ ■■■■■)

Project ko run karne ke liye nimn steps follow kiye jaate hain:

Step 1: Open Terminal / Shell: cd "d:\student management project"

Step 2: Activate Virtual Environment: .\venv\Scripts\activate

Step 3: Run Application: python app.py (Yeh command database tables aur default admin auto-create kar degi).

Step 4: Access in Browser: Open Chrome/Edge and navigate to **http://127.0.0.1:5000**

Step 5: Sign In: Username: admin | Password: admin123

9. Conclusion & Future Enhancements (■■■■■■■■ ■■ ■■■■■■ ■■ ■■■■■)

Conclusion: EduManage ek robust, scalable aur user-friendly Student Management System hai jo educational institutions ke operational workflow ko streamline karta hai. Iska modern Glassmorphism UI, client-side live search, aur automated calculations is project ko standard student projects se alag aur commercial-grade banate hain.

Future Scope:

- **Student & Parent Portal:** Separate student login jaha ve apna attendance record aur fee receipt download kar sakein.
- **Automated SMS / Email Alerts:** Absent students ke parents ko automatic SMS / Email notification.
- **Excel / CSV Bulk Export & Import:** Bulk student admissions aur exam score sheets ko Excel format me download karna.
- **Online Payment Gateway:** Razorpay / UPI integration for cashless online fee payment.

Project Status: Production Ready (Phase 4 Completed)	**Documentation Version:** 1.0.0 (Final)	**Report Compiled:** 18-10-2026